Name:

Grade: ~~200~~ 0 /100   IG

Linux Server & Bash Programming — Final Exam

2025-01-07

Instructions

1. **No computers, no phones, no notes**
2. **No talking with other people in the room**
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Grade* box!)
5. For each question, you'll need to:
   - answer the question by crossing with an `X` the right answer
   - give an estimation of your confidence in the answer given (`1-10`)
6. Once you're done:
   - Double-check your answers
   - Hand over your test to me
   - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

Each question gives up to **10 points**. There are **10 questions**, for a total of maximum **100 points**.

Each correct answer gives **1** point, each incorrect answer gives **-1** point. The point from the answer is **multiplied by your confidence level**, for a range of -10 to 10 points per answer.

	Confidence 1/10	**Confidence 6/10**	**Confidence 10/10**
Correct answer	1 point	6 points	10 points
Incorrect answer	-1 point	-6 points	-10 points

Grading

Points	Grade
-100 – 50	IG
51 – 89	G
90 – 100	VG

-5

1 In Linux everything is. . .

I'm ~~10~~ 5 */10 confident about my answer*

1. Difficult
2. More fun
3. A file
4. A terminal
5. A shell v

-5

2 I have added new storage to my linux server. In which order do I need to create everything needed for an encrypted filesystem?

I'm 5 */10 confident about my answer*

1. Block device -> partitions -> LUKS encryption -> filesystem
2. Partitions -> filesystem -> block device -> LUKS encryption
3. LUKS encryption -> block device -> filesystem -> partitions
4. LUKS encryption -> block device -> partitions -> filesystem

3 How many hosts can I have in a /27 network? (NB: hosts, not addresses)

0

I'm */10 confident about my answer*

1. 2^4
2. $2^4 - 2$
3. 2^5
4. $2^5 - 2$
5. 2^6
6. $2^6 - 2$

-10

4 How can I access another machine's `localhost`?

I'm 10 */10 confident about my answer*

1. By adding an entry to `/etc/hosts`
2. By creating or modifying a file in `/etc/netplan/`
3. It's not possible
4. By opening a listening socket in the target machine and accepting connections

5 DNS is...

+10

I'm 10 */10 confident about my answer*

1. A service that translates IP addresses into hostnames
2. A service that translates hostnames into IP addresses

6 Bash stands for...

10

I'm 10 */10 confident about my answer*

1. Better Ask for Some Help
2. Better than Any other SHell
3. Bourne Again SHell
4. Building Advanced Scripts Hub

7 To link together two processes, passing the output of one as the input for the next, we use...

5

I'm 5 */10 confident about my answer*

1. `|`
2. `>`
3. `>>`
4. `<`
5. `<<`
6. `2&>`

8 I'm writing a bash script, and I want to process the given cli arguments one by one, removing them from the argument list as I do so. To achieve that, I use...

-10

I'm 10 */10 confident about my answer*

1. `STDERR`
2. `STDIN`
3. `STDOUT`
4. `case`
5. `esac`
6. `shift`
7. `unshift`

S

9 A docker container is...

I'm 8 */10 confident about my answer*

1. Like a variable, but global
2. Like a script, but better
3. A kind of virtual machine
4. An isolated process

4

10 In docker there are two types of volumes we can use, ...

∅

I'm */10 confident about my answer*

1. Isolated volumes and shared volumes
2. Unbound volumes and unnamed volumes
3. Bind volumes and named volumes
4. It's a trick question, there is only 1 kind of volumes